Document Summary
This note explores the evolution of the GPT family of models and the broader trends in large language models (LLMs). It covers the progression from GPT-1 to GPT-4, the impact of scaling laws, the rise of larger models, the shift toward more efficient training, and the emergence of open-source alternatives like the Llama family.

GPT Family Evolution and LLM Trends

The GPT Family

Evolution of GPT Models

- **GPT-1 (2018)**
  - First in a (long series) of models
  - Decoder-only transformer architecture
  - Pretraining on a large corpus, fine-tuning on various tasks
  - **Architecture**:
    - 12 layers, decoder-only
    - 12 heads each
    - 768 dimensional states
    - BPE with 40,000 merges
    - Learned positional embeddings
    - Context size: 512 tokens
    - 117M parameters
  - **Training**:
    - Unsupervised pretraining on BooksCorpus (7,000 books, ~5 GB of text)
    - Fine-tuning tasks: Natural language inference, question answering, semantic similarity, classification
- **GPT-2 (2019)**
  - Larger training set
  - Initial controversies around the (lack of) release for potential misuse
  - **Setup**:

    - Unsupervised pretraining on WebText (45M links, 40GB of text)
    - No fine-tuning
    - BPE with 50,257 tokens
    - Context size: 1024
    - Up to 1.5B parameters
- **GPT-3 (2020)**
  - Scaling up: 175B parameters
  - No other meaningful architectural changes w.r.t. GPT-2
  - **Training**:
    - Multiple datasets: (Filtered) CommonCrawl, WebText2, Books1, Books2, Wikipedia
    - No fine-tuning done on GPT-3 – in-context learning only
  - **In-context performance**:
    - Larger models show remarkable 1- and few-shot performance
    - The gap between 0- and 1-shot performance is remarkable
- **GPT-4 (2023)**
  - OpenAI no longer providing information on the model

Scaling Laws for Neural Language Models

Published in January 2020, by OpenAI

Shows various empirical takeaways

The loss scales as a power-law with model size, dataset size, and amount of compute

Other Takeaways

- Performance depends very weakly on other architectural hyperparameters such as depth vs. width (number of layers vs embedding size) for a fixed overall number of parameters
- Empirically optimal results:
  - $N \propto C^{0.54}$, $B \propto C^{0.35}$, $S \propto C^{0.07}$
  - Where N = model size, B = batch size, S = number of steps, C = computing budget

The Race to Bigger Models

These scaling laws resulted in a race toward building larger models

GPT-3 was the first 100B+ parameters model

Other larger models have been developed, following this trend

Big Models Got Bigger

- **Jurassic-1**: 178B parameters, AI21labs (2021)
- **Gopher**: 280B parameters, DeepMind (2021)
- **Megatron-Turing NLG**: 530B parameters, NVIDIA + Microsoft (2021)
- **PaLM (Pathways Language Model)**: 540B parameters, Google (2022)

Oversized and Undertrained!

DeepMind publishes "Training Compute-Optimal Large Language Models" in March 2022

Main claims:

- Current large language models are significantly under-trained
- For every doubling of model size, the number of training tokens should also be doubled
- Chinchilla, a new correctly-sized model, outperforms larger ones

Chinchilla

- **70B parameters** model
- Trained on the same compute budget as Gopher (280B)
- $5.76 \cdot 10^{22}$ FLOPs
- Chinchilla is ¼ of Gopher's size, and is trained on more tokens (1.4T vs 300B)
- Chinchilla generally outperforms Gopher, but also GPT-3, on various tasks

A New Trend

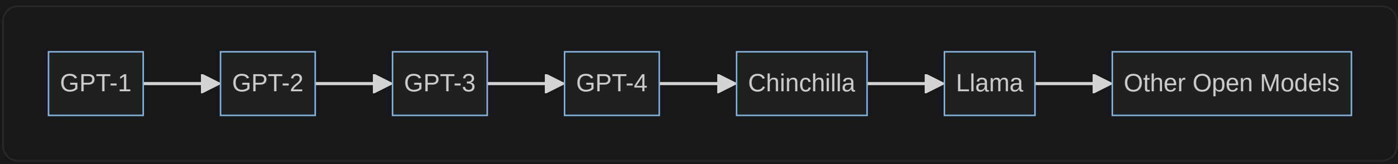
- The previous trend of "always larger" models started fading
- There has since been a return to smaller models, trained for longer
- Smaller models can achieve better performance!
- LLMs become more accessible
- This led to a large ecosystem of (open) models

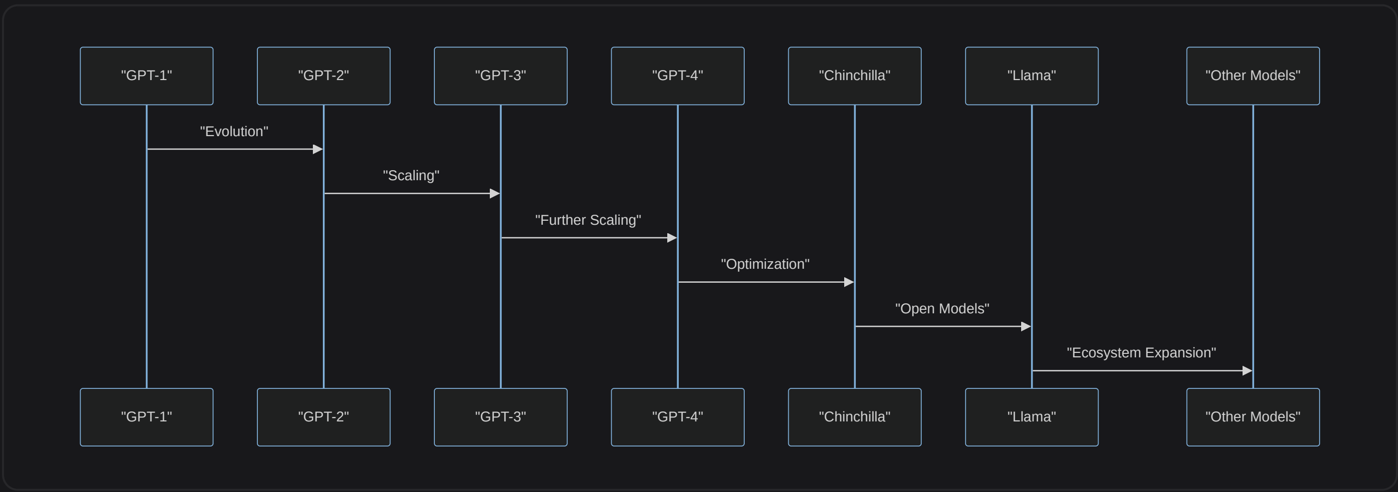
Llama Family

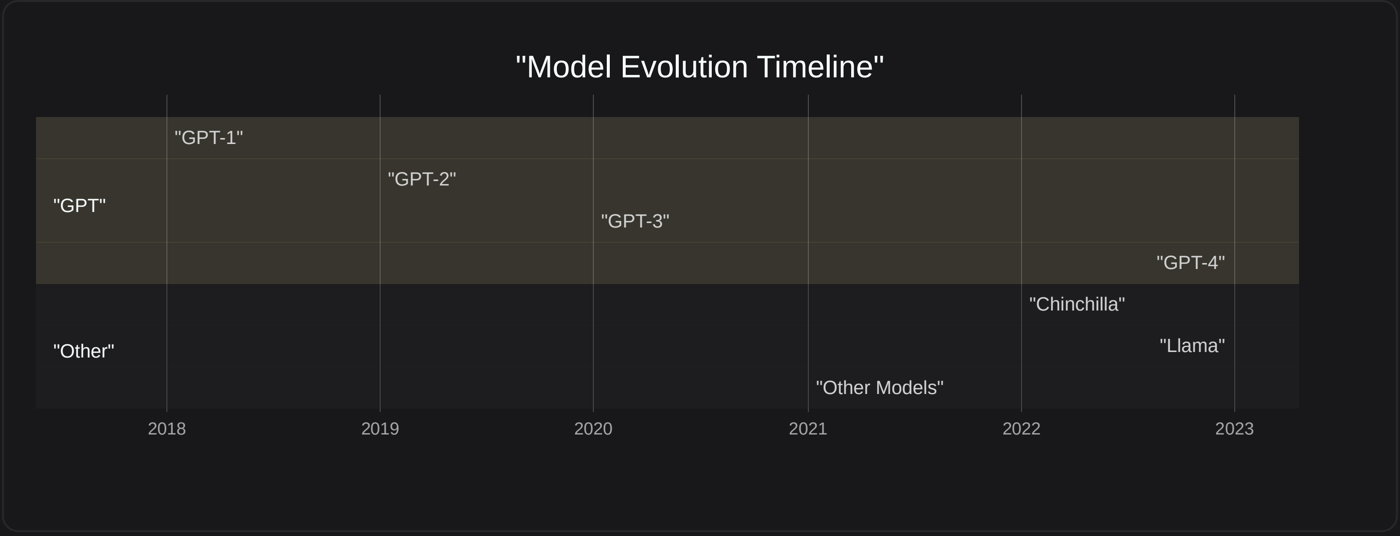
- **Llama (Large Language Model Meta AI)** is a family of models introduced by Meta AI, starting in 2023
- All autoregressive, decoder-only architectures, trained on open datasets
- All models are openly available
- **Versions**:
  - LLaMA (Feb '23): 7B, 13B, 32B, 65B
  - Llama 2 (Jul '23): 7B, 13B, 70B
  - Llama 3 (Apr '24): 8B, 70B
  - 3.1 (Jul '24): 8B, 70B, 405B
  - 3.2 (Sep '24): 1B, 3B, 11B, 90B (with multimodal version)
  - Plus other versions (e.g., Code Llama – based on Llama 2, or instruction-tuned models)

Other Families of Open Models

- **GPT-Neo/GPT-J** (EleutherAI, 🇺🇸) – open source alternatives to the GPT family
- **Mistral** (MistralAI, 🇫🇷) – wide variety of model sizes, code-tuned versions (for 80+ languages), multimodal versions (Pixtral)
- **GLM** (Zhipu AI, 🇨🇳) – General Language Model, more oriented toward the Chinese language, but also works well on other languages, including English
- **Falcon** (Technology Innovation Institute, 🇦🇪) – different sized models, they also released a Mamba-based model (State Space Language Models!)







"Model Evolution Timeline"
"GPT"
"GPT-1"
"GPT-2"
"GPT-3"
"GPT-4"
"Other"
"Chinchilla"
"Llama"
"Other Models"
2018
2019
2020
2021
2022
2023